

Project | Sustainability Impact Analysis for Intel



Introduction

Intel is committed to reducing its carbon footprint by improving the sustainability of its products throughout their entire lifecycle. One of the company's key sustainability initiatives is its **device repurposing program**, which extends the lifespan of electronic devices through refurbishment and recycling, reducing electronic waste and conserving valuable resources.

This project analyzes Intel's 2024 device repurposing dataset using SQL to evaluate the effectiveness of the company's sustainability initiatives. By examining environmental performance across different device categories and sustainability programs, the analysis identifies key trends and opportunities for improvement.

The primary objective of this project is to provide data-driven insights that support strategic decision-making and help optimize Intel's sustainability efforts while maximizing environmental impact.

Project Objectives

This project aims to answer the following business questions:

- Which device categories generate the greatest environmental impact?
 - Which sustainability initiatives produce the highest overall benefits?
 - How do environmental performance metrics vary across different programs?
 - What data-driven recommendations can improve Intel's sustainability strategy?
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Technical Skills

- SQL
- Data Analysis
- Data Cleaning
- Aggregate Functions
- JOIN Operations
- Common Table Expressions (CTEs)
- Window Functions
- Business Intelligence
- Sustainability Analytics

Dataset Overview

This project analyzes two relational datasets, **device_data** and **impact_data**, which are joined using the **device_id** field to evaluate Intel's device repurposing strategy and its environmental sustainability impact.

`intel.device_data`

- **device_id**: Unique identifier for each repurposed device.
- **device_type**: Device category, either **Laptop** or **Desktop**
- **model_year**: Manufacturing year of the device (e.g., 2018, 2019, etc.).

`intel.impact_data`

- **impact_id**: Unique identifier for each sustainability impact record (e.g., "LP20NA141592").
- **device_id**: Foreign key linking each impact record to the corresponding device in the **intel.device_data** table.
- **usage_purpose**: Primary purpose of device repurposing. Values include **Education & Digital Literacy**, **Corporate & Enterprise**, **Government & Public Sector**, **Environmental Sustainability Programs**, and **Social Impact & Non-Profit**.
- **power_consumption**: Device power consumption during operation, measured in watts (W).

- **energy_savings_yr**: Estimated annual energy savings achieved by repurposing the device instead of manufacturing a new one, measured in kilowatt-hours (kWh).
- **co2_saved_kg_yr**: Estimated annual CO2 emissions avoided through device repurposing, measured in kilograms (kg).
- **recycling_rate**: Percentage of the device that can be recycled.
- **region**: Geographic region where the device was repurposed. Values include **North America**, **Europe**, and **Asia**.

– Data Preparation and Feature Engineering

The first step was to combine the device and environmental impact datasets using the shared **device_id** field. This created a unified dataset containing device characteristics, repurposing regions, energy savings, and CO2 reduction metrics.

Joining the Datasets

The following query uses an INNER JOIN to combine matching records from intel.device_data and intel.impact_data

```
SELECT *  
FROM intel.device_data d  
INNER JOIN intel.impact_data i  
    ON d.device_id = i.device_id;
```

Calculating Device Age

A new field, **device_age**, was calculated by subtracting the device's manufacturing year from the analysis year, 2024.

(paste your query below 📌)

```
SELECT  
    d.*,  
    i.*,
```

```
2024 - d.model_year AS device_age
FROM intel.device_data d
INNER JOIN intel.impact_data i
ON d.device_id = i.device_id;
```

Initial Observation

After ordering the dataset by **model_year** from oldest to newest, the analysis shows that **newer devices make up the largest share of repurposed devices**. This suggests that Intel prioritizes repurposing relatively modern devices because they remain functional for a longer period and can continue providing value after refurbishment. This strategy may also improve sustainability outcomes while reducing replacement costs.

Creating Device Age Categories

To support clearer comparisons, devices were grouped into three age categories:

- **newer:** 3 years old or less
- **mid-age:** more than 3 years old and up to 6 years old
- **older:** more than 6 years old

```
SELECT
  d.*,
  i.*,
  2024 - d.model_year AS device_age,
  CASE
```

```
        WHEN 2024 - d.model_year <= 3 THEN 'newer'
        WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
        ELSE 'older'
    END AS device_age_bucket
FROM intel.device_data d
INNER JOIN intel.impact_data i
    ON d.device_id = i.device_id;
```

– Key Performance Metrics

To evaluate the overall effectiveness of Intel's device repurposing program, key performance indicators (KPIs) were calculated using the cleaned and prepared dataset created in the previous section. These metrics provide a high-level overview of the program's scale, environmental impact, and sustainability performance.

Total Devices Repurposed

Total Devices Repurposed in 2024

601740

The dataset represents every device repurposed by Intel during 2024, providing a comprehensive view of the company's sustainability initiatives.

```
WITH device_data AS (
    SELECT
        d.*,
        i.*,
        2024 - d.model_year AS device_age,
        CASE
            WHEN 2024 - d.model_year <= 3 THEN 'newer'
```

```
        WHEN 2024 - d.model_year <= 6 THEN
        'mid-age'
        ELSE 'older'
        END AS device_age_bucket
    FROM intel.device_data d
    INNER JOIN intel.impact_data i
        ON d.device_id = i.device_id
    )

    SELECT
        COUNT(*) AS total_devices_repurposed,
        AVG(device_age) AS avg_device_age,
        AVG(energy_savings_yr) AS avg_energy_savings_kwh,
        SUM(co2_saved_kg_yr) / 1000 AS total_co2_saved_tons
    FROM device_data;
```

Environmental Impact Analysis

Intel's repurposing program saved an average of 25.74 kWh of energy per device each year and prevented approximately 6,768 tons of CO₂ emissions annually. The CO₂ savings are especially significant because they are roughly equivalent to removing more than 1,400 passenger cars from the road for a year. While the energy savings per device may seem small, the impact becomes substantial when multiplied across thousands of repurposed devices. These results show that extending the life of existing devices can reduce e-waste, lower manufacturing demand, and create meaningful environmental benefits.

– Trend Analysis & Sustainability Optimization

This section explores how different device characteristics influence Intel's sustainability performance. By grouping devices according to **device type**, **device age**, and **geographic**

region, the analysis identifies which categories generate the greatest environmental impact and where Intel can maximize the effectiveness of its repurposing strategy. .

Analysis by Device Type

The following query compares the total number of repurposed devices, average annual energy savings, and average annual CO₂ emissions avoided for each device type.

```
WITH device_data AS (  
    SELECT  
        d.*,  
        i.*,  
        2024 - d.model_year AS device_age,  
        CASE  
            WHEN 2024 - d.model_year <= 3 THEN 'newer'  
            WHEN 2024 - d.model_year <= 6 THEN  
'mid-age'  
            ELSE 'older'  
        END AS device_age_bucket  
    FROM intel.device_data d  
    INNER JOIN intel.impact_data i  
        ON d.device_id = i.device_id  
)  
  
SELECT  
    device_type,  
    COUNT(*) AS total_devices,  
    AVG(energy_savings_yr) AS avg_energy_savings_kwh,  
    AVG(co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons  
FROM device_data  
GROUP BY device_type;
```

Business Insight

Laptops contribute the most to energy savings and CO₂ reduction. Although laptops and desktops have similar average energy savings and CO₂ reductions per device, laptops perform slightly better in both categories. Additionally, Intel repurposed far more laptops (408,064) than desktops (193,676), which increases their overall environmental impact. This may be because laptops are more common, consume less power, and remain useful for a longer period when repurposed. Therefore, focusing on laptop repurposing can help maximize sustainability benefits and reduce overall emissions.

Analysis by Device Age

The following query compares sustainability performance across different device age groups.

```
WITH device_data AS (  
  SELECT  
    d.*,  
    i.*,  
    2024 - d.model_year AS device_age,  
    CASE  
      WHEN 2024 - d.model_year <= 3 THEN 'newer'  
      WHEN 2024 - d.model_year <= 6 THEN  
'mid-age'  
      ELSE 'older'  
    END AS device_age_bucket  
  FROM intel.device_data d  
  INNER JOIN intel.impact_data i
```



```
        ON d.device_id = i.device_id
    )

    SELECT
        device_age_bucket,
        COUNT(*) AS total_devices,
        AVG(energy_savings_yr) AS avg_energy_savings_kwh,
        AVG(co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons
    FROM device_data
    GROUP BY device_age_bucket;
```

Business Insight

The results suggest that newer devices make up a larger share of the repurposed devices, while older devices tend to generate higher average energy savings per device. This indicates that although Intel repurposes more newer devices, repurposing older devices may provide greater environmental benefits on a per-device basis because replacing older, less efficient technology can save more energy. Therefore, there is a tradeoff between the number of devices repurposed and the amount of energy saved by each device.

Analysis by Region

The following query evaluates regional differences in device repurposing performance.

```
WITH device_data AS (
    SELECT
        d.*,
        i.*,
```

```

        2024 - d.model_year AS device_age,
        CASE
            WHEN 2024 - d.model_year <= 3 THEN 'newer'
            WHEN 2024 - d.model_year <= 6 THEN
'mid-age'
            ELSE 'older'
        END AS device_age_bucket
    FROM intel.device_data d
    INNER JOIN intel.impact_data i
        ON d.device_id = i.device_id
)

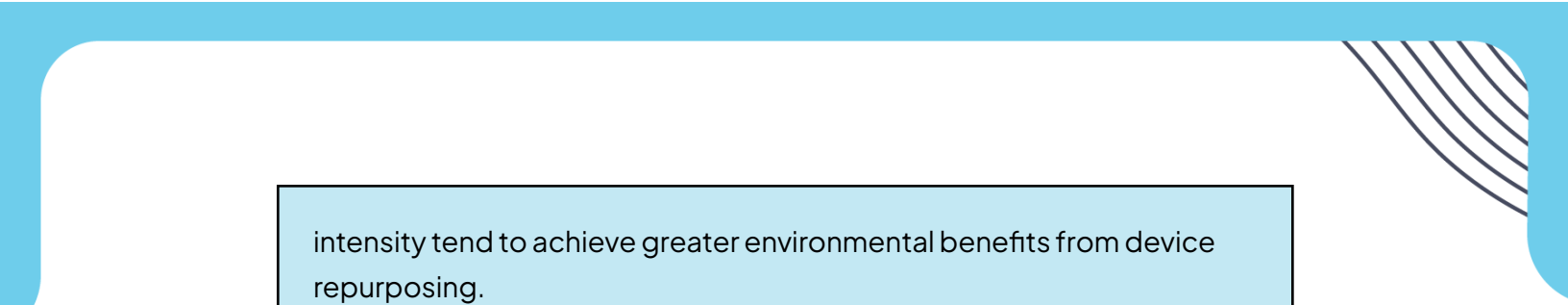
SELECT
    region,
    COUNT(*) AS total_devices,
    AVG(energy_savings_yr) AS avg_energy_savings_kwh,
    AVG(co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons
FROM device_data
GROUP BY region;

```

Business Insight

Write your answer here.

The results suggest that carbon intensity has a significant impact on CO₂ savings from repurposed devices. Asia has the highest average CO₂ savings per device (0.0155 tons), followed by North America (0.0103 tons), while Europe has the lowest (0.0064 tons). This indicates that repurposing devices in Asia provides the greatest environmental benefit. One possible reason is that electricity generation in Asia is generally more carbon-intensive, meaning that avoiding the production of new devices prevents more emissions. Europe likely has lower CO₂ savings because its electricity grid relies more on renewable and low-carbon energy sources. Therefore, regions with higher carbon



intensity tend to achieve greater environmental benefits from device repurposing.

– Strategic Recommendations

The analysis provides several important insights into Intel's device repurposing strategy. Based on the observed trends in device age, energy savings, CO₂ reduction, and regional performance, the following recommendations are proposed to maximize the environmental impact of Intel's sustainability initiatives.

Key Findings

Newer and mid-age devices account for the majority of Intel's repurposed devices, demonstrating the company's emphasis on extending the lifespan of relatively modern technology.

Older devices generate greater environmental benefits on a per-device basis, as replacing older and less energy-efficient hardware results in larger energy savings and greater reductions in CO₂ emissions.

Laptops contribute the greatest overall sustainability impact because they represent the majority of repurposed devices while also delivering slightly higher average energy savings and CO₂ reductions than desktop computers.

Regional differences significantly influence environmental outcomes. Asia achieves the highest average CO₂ reduction per repurposed device, suggesting that repurposing efforts produce greater environmental benefits in regions with higher electricity carbon intensity.

Strategic Recommendation

Based on the analysis, Intel should prioritize repurposing laptops while expanding refurbishment efforts for older devices in regions with higher carbon intensity, particularly Asia.

Laptops represent the largest share of repurposed devices and consistently provide strong environmental performance. Although newer devices dominate the current repurposing program, older devices deliver greater energy savings and CO₂ reductions on a per-device basis because replacing less efficient technology avoids more emissions. Furthermore, the analysis shows that Asia achieves the highest average CO₂ reduction per device, making it the region where additional repurposing efforts can generate the greatest environmental impact.

By allocating more resources toward high-impact device categories, older equipment, and carbon-intensive regions, Intel can maximize energy savings, reduce greenhouse gas emissions, and further strengthen its long-term sustainability strategy.

Project Reflection

ChatGPT helped me connect the data to a practical business recommendation. I initially focused on the differences between device types, but ChatGPT helped me consider the impact of device age and regional carbon intensity as well. This improved my recommendation by making it more data-driven and strategic. The suggestion to prioritize older devices and focus on regions such as Asia, where CO₂ savings are highest, helped strengthen the overall sustainability impact of the recommendation.

– Advanced Strategy Optimization

To further optimize Intel's sustainability strategy, the analysis compares the percentage of total energy savings and CO2 reductions contributed by each device type within each geographic region. This provides deeper insight into where Intel's repurposing resources can generate the greatest environmental impact.

```
WITH device_data AS (  
    SELECT  
        d.*,  
        i.*,  
        2024 - d.model_year AS device_age,  
        CASE  
            WHEN 2024 - d.model_year <= 3 THEN 'newer'  
            WHEN 2024 - d.model_year <= 6 THEN  
'mid-age'  
            ELSE 'older'  
        END AS device_age_bucket  
    FROM intel.device_data d  
    INNER JOIN intel.impact_data i  
        ON d.device_id = i.device_id  
) ,  
  
region_totals AS (  
    SELECT  
        region,  
        SUM(energy_savings_yr) AS  
region_energy_savings,  
        SUM(co2_saved_kg_yr) AS region_co2_savings  
    FROM device_data  
    GROUP BY region  
)  
  
SELECT  
    d.region,
```

```

        d.device_type,
        COUNT(*) AS total_devices,
        AVG(d.energy_savings_yr) AS avg_energy_savings_kwh,
        AVG(d.co2_saved_kg_yr) / 1000 AS
avg_co2_saved_tons,

        (SUM(d.energy_savings_yr) /
r.region_energy_savings) * 100
        AS pct_energy_savings,

        (SUM(d.co2_saved_kg_yr) / r.region_co2_savings) *
100
        AS pct_co2_reduction

FROM device_data d
JOIN region_totals r
    ON d.region = r.region

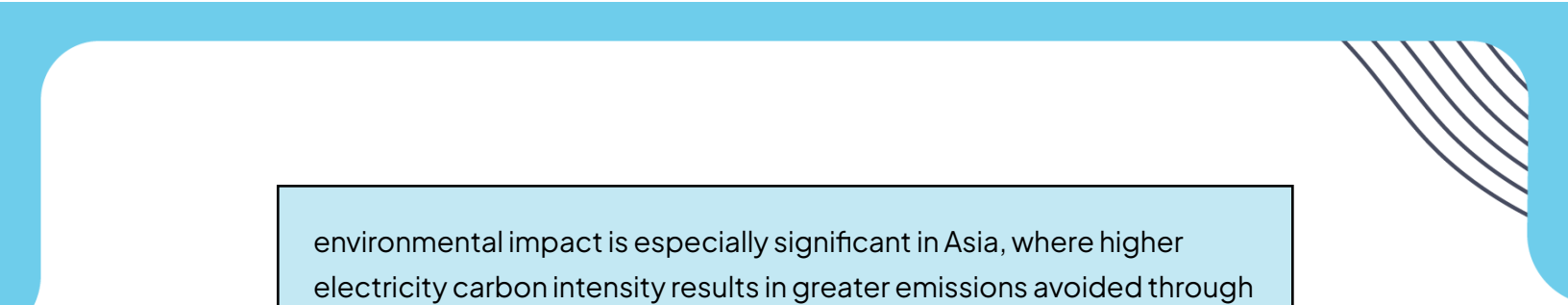
GROUP BY
    d.region,
    d.device_type,
    r.region_energy_savings,
    r.region_co2_savings

ORDER BY
    d.region,
    d.device_type;

```

Strategic Analysis

The analysis shows that laptops contribute the largest share of both energy savings and CO₂ reductions across most regions. Their



environmental impact is especially significant in Asia, where higher electricity carbon intensity results in greater emissions avoided through device repurposing.

These findings suggest that Intel can maximize environmental benefits by prioritizing laptop repurposing initiatives in Asia and other high-impact regions. Allocating additional resources to the device types and geographic areas that contribute the greatest share of sustainability gains will improve the overall effectiveness of Intel's environmental strategy.

Cost Optimization Strategy

While environmental impact is a primary objective, long-term sustainability programs should also consider cost-effectiveness.

To incorporate financial analysis, a **cost_per_device** field could be added to the existing dataset or joined from a separate cost table. This would enable additional performance metrics such as:

- Cost per kWh of energy saved
- Cost per metric ton of CO₂ reduced
- Return on sustainability investment (ROSI)

These metrics would help identify the device categories and regions that generate the greatest environmental benefits at the lowest operational cost.

Intel could further optimize its repurposing strategy by prioritizing devices that provide high energy savings and significant CO₂ reductions while requiring relatively low refurbishment costs. For example, if laptops repurposed in Asia deliver the highest



environmental return at a competitive cost, they should receive greater investment. Conversely, devices with high refurbishment costs but limited environmental benefits should receive lower priority.

Balancing sustainability performance with cost efficiency enables Intel to maximize environmental impact while making the most effective use of available resources.